Noise

Quantum Noise Simulation

A step-by-step chapter: from density matrices to noisy circuits in CUDA-Q

What you will be able to do after this chapter

- compute with density matrices, Bloch vectors, purity and fidelity without sign or factor-of-2 errors;
- write down and apply the Kraus operators of the standard channels and check that a map is CPTP;
- relate amplitude and phase damping to T_1, T_2 and T_φ, and model readout errors;
- distinguish coherent errors, stochastic noise, decoherence and measurement errors, and estimate how they accumulate;
- read the noise model and the results of `Noise_Simulation.ipynb`, and choose a CUDA-Q simulation method;
- explain zero-noise extrapolation, readout mitigation, dynamical decoupling and the bit-flip code — and what each can and cannot do;
- solve every type of question in the accompanying 80-question assessment.

SHIRO FELLER

Companion material in the same repository: `presentations/Noise_Lecture.pdf` (slides), `Noise_Simulation.ipynb` (CUDA-Q notebook) and `assessments/` (question paper and answer book).

Contents

Quantum Noise Simulation **3**

1 Error statistics . . . 3
2 Density matrices and the Bloch ball . . . 4
3 Decoherence and fidelity . . . 4
4 Quantum channels and CPTP maps . . . 4
5 Pauli channels . . . 5
6 Amplitude damping, phase damping, T_1 and T_2 . . . 5
7 Measurement errors . . . 6
8 Coherent and stochastic errors . . . 6
9 Simulating noise with CUDA-Q . . . 6
10 The notebook and its results . . . 7
11 Fidelity measures and benchmarking . . . 8
12 Error mitigation . . . 8
13 Dynamical decoupling and error correction . . . 8
14 Problem-solving strategies and question map . . . 9
15 Practice problems . . . 9

SILICO FELLER

Quantum Noise Simulation

How to use this chapter

Sections 1–3 are the *mathematics* (error statistics, density matrices, decoherence and fidelity). Sections 4–8 are the *noise models* (channels, Pauli noise, damping and T_1/T_2, readout, coherent errors). Sections 9–10 are the *simulation* (CUDA-Q methods and the notebook). Sections 11–13 are the *advanced* material (fidelity measures and benchmarking, mitigation, decoupling and error correction). Section 14 has strategies and a map from every assessment question to its section; Section 15 has practice problems with answers.

Statements are marked as *implemented in the repository*, *theory only* or *illustrative experiment* (run for the lecture with `presentations/noise_source/make_figures.py`). Everything was simulated on a CPU; no quantum processor was used.

Definition: Conventions used everywhere in this chapter

- Qubit 0 is written first; $\rho = \frac{1}{2}(I + r_x X + r_y Y + r_z Z)$; $\langle Z \rangle = P(0) - P(1)$; $F = \langle\psi|\rho|\psi\rangle$.
- Channel $\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger$, $\sum_k K_k^\dagger K_k = I$.
- Depolarizing in CUDA-Q's convention: $\sqrt{1-p}I$, $\sqrt{p/3}\{X, Y, Z\}$; amplitude damping $\mathrm{diag}(1, \sqrt{1-\gamma})$, $\sqrt{\gamma}\,|0\rangle\langle 1|$; phase damping $\mathrm{diag}(1, \sqrt{1-\lambda})$, $\mathrm{diag}(0, \sqrt{\lambda})$.
- `Noise_Simulation.ipynb`: 15-qubit GHZ, channel with probability p after every `h`, `x`, `ry`, `rz` and CNOT; no readout noise; 10 000 shots, seed 42; recorded on `stim` (CPU).

1 Error statistics

Key result: independent errors

G independent locations with error probability p: $P(\text{no error}) = (1-p)^G \approx e^{-pG}$, $P(\geq 1) = 1 - (1-p)^G$, $P(\text{exactly } k) = \binom{G}{k} p^k (1-p)^{G-k}$. Shot noise of an estimated probability: $\sqrt{P(1-P)/S}$.

Worked example 1.1: a 20-gate circuit and 10 000 shots

$0.99^{20} = 0.818$, so an error occurs in 18.2 % of runs ($20p = 20\,\%$ over-counts). A probability of 0.9 from 10^4 shots: ± 0.003.

Common trap

Adding error probabilities (Gp) instead of multiplying survival probabilities; forgetting the square root in a standard error.

2 Density matrices and the Bloch ball

Definition: density matrix

$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$: Hermitian, trace 1, positive semi-definite. For a qubit $\rho = \frac{1}{2}(I + \vec{r}\cdot\vec{\sigma})$, $|\vec{r}| \le 1$, $r_z = \rho_{00} - \rho_{11}$, $r_x + ir_y = 2\rho_{10}$; purity $\operatorname{Tr}\rho^2 = \frac{1+|\vec{r}|^2}{2}$; eigenvalues $\frac{1\pm|\vec{r}|}{2}$.

Worked example 2.2: reading a density matrix

$\rho = \begin{pmatrix} 0.8 & 0.2-0.1i \\ 0.2+0.1i & 0.2 \end{pmatrix}$: $\vec{r} = (0.4, 0.2, 0.6)$, purity $\frac{1+0.56}{2} = 0.78$. $\begin{pmatrix} 0.6 & 0.5 \\ 0.5 & 0.4 \end{pmatrix}$ is not a state ($\det < 0$). $X\rho X$ maps $(r_x, r_y, r_z) \to (r_x, -r_y, -r_z)$.

Common trap

Forgetting the complex conjugate in $|\psi\rangle\langle\psi|$; reading r_y from ρ_{01} with the wrong sign; taking diagonal entries for eigenvalues.

3 Decoherence and fidelity

A qubit that becomes entangled with its environment, $\alpha|0\rangle|e_0\rangle + \beta|1\rangle|e_1\rangle$, has coherence multiplied by $\langle e_1|e_0\rangle$. Random phases average to the same effect: φ uniform on $[-\frac{\pi}{2}, \frac{\pi}{2}]$ gives $\overline{\langle X\rangle} = \frac{2}{\pi}$.

Key result: fidelity

$F = |\langle\psi|\phi\rangle|^2$ (pure), $F = \langle\psi|\rho|\psi\rangle$ (pure target), $F(|+\rangle, \rho) = \frac{1+r_x}{2}$. GHZ: $F = \frac{1}{2}(\rho_{0\cdots0} + \rho_{1\cdots1}) + \operatorname{Re}\rho_{0\cdots0,1\cdots1}$. Mixing with white noise: $F = (1-q) + q/d$.

Worked example 3.3: reduced states are mixed

$\sqrt{0.8}|00\rangle + \sqrt{0.2}|11\rangle$: $\rho_A = \operatorname{diag}(0.8, 0.2)$, purity 0.68.

4 Quantum channels and CPTP maps

Definition: channel

A linear, completely positive, trace-preserving map. Kraus form $\mathcal{E}(\rho) = \sum K\rho K^\dagger$ guarantees CP; $\sum K^\dagger K = I$ is trace preservation. Kraus sets are not unique; at most d^2 operators are needed. Choi state $J = (\mathcal{E}\otimes\mathrm{id})(|\Phi^+\rangle\langle\Phi^+|)$; process fidelity $\langle\Phi^+|J|\Phi^+\rangle$.

Method: checking and applying a channel

Step 1. Compute $\sum K^\dagger K$; it must be I (e.g. $\operatorname{diag}(1, 0.8)$ and $b|0\rangle\langle 1|$ need $b = 0.6$).

Step 2. Apply term by term: bit flip $p = 0.2$ on $\begin{pmatrix} 0.9 & 0.3 \\ 0.3 & 0.1 \end{pmatrix}$ gives $\begin{pmatrix} 0.74 & 0.30 \\ 0.30 & 0.26 \end{pmatrix}$.

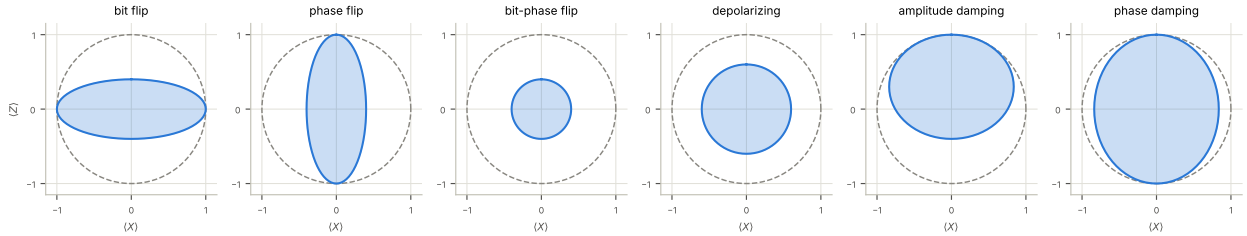
Step 3. Compose channels by composing Bloch maps: depolarizing 0.03 then 0.06: $0.96 \times 0.92 = 0.8832 \Rightarrow p = 0.0876$.

5 Pauli channels

channel	Kraus	Bloch map
bit flip	$\sqrt{1-p}I$, $\sqrt{p}X$	(r_x, cr_y, cr_z), $c = 1-2p$
phase flip	$\sqrt{1-p}I$, $\sqrt{p}Z$	(cr_x, cr_y, r_z)
bit-phase flip	$\sqrt{1-p}I$, $\sqrt{p}Y$	(cr_x, r_y, cr_z)
depolarizing	$\sqrt{1-p}I$, $\sqrt{p/3}\{X,Y,Z\}$	$(1-\frac{4p}{3})\vec{r}$

Worked example 5.4: flip probabilities

Depolarizing $p = 0.03$ on $|0\rangle$: $P(1) = \frac{2p}{3} = 0.02$. Two-qubit depolarizing: the control's Z outcome flips with $\frac{8p}{15}$. Phase flip $p = 0.1$ on $|+\rangle$: $P(+) = 0.9$, $\langle X\rangle = 0.8$. Phase flip with $p = 1$ is the unitary Z — reversible.



x–z section of the Bloch ball (dashed) and its image for $p = \gamma = \lambda = 0.3$: bit flip, phase flip, bit-phase flip, depolarizing, amplitude damping, phase damping (illustrative experiment, checked with CUDA-Q).

Key result: phase errors are invisible in the Z basis

For a GHZ state with phase-flip noise the Z population stays 1 while $\langle X^{\otimes N}\rangle$ drops (lecture: 0.0504 at $p = 0.05$, 15 qubits, `stim`).

6 Amplitude damping, phase damping, T_1 and T_2

Key result: damping channels

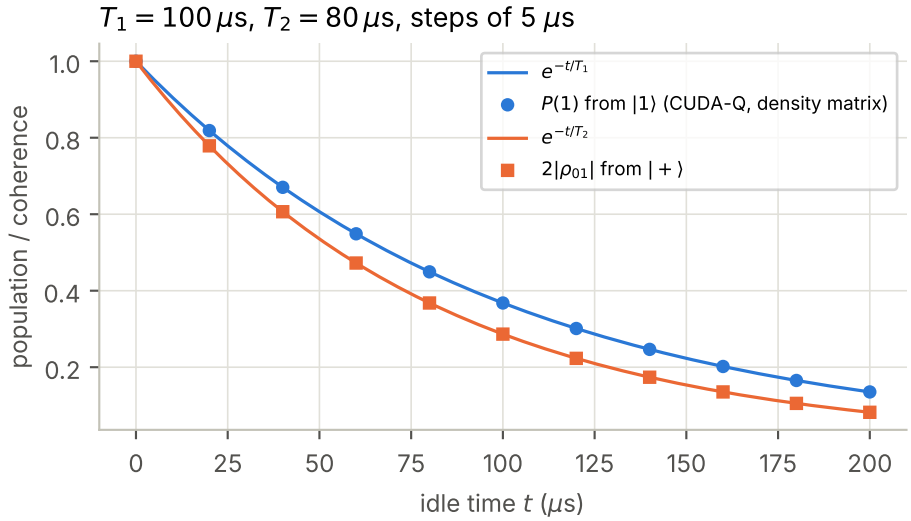
Amplitude damping: $\rho_{11} \to (1-\gamma)\rho_{11}$, $\rho_{00} \to \rho_{00}+\gamma\rho_{11}$, $\rho_{01} \to \sqrt{1-\gamma}\rho_{01}$; non-unital. Phase damping: $\rho_{01} \to \sqrt{1-\lambda}\rho_{01}$, populations fixed; equal to a phase flip with $1-2p = \sqrt{1-\lambda}$.

Key result: relaxation times

$\rho_{11}(t) = \rho_{11}(0)e^{-t/T_1}$; $|\rho_{01}(t)| = |\rho_{01}(0)|e^{-t/T_2}$; $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\varphi}$, so $T_2 \le 2T_1$. Per step Δt: $\gamma = 1-e^{-\Delta t/T_1}$, $\sqrt{1-\lambda} = e^{-\Delta t/T_\varphi}$.

Worked example 6.5: an idling qubit

$T_1 = 80$, $T_2 = 60\,\mu$s, $|+\rangle$, $t = 20\,\mu$s: $\rho_{11} = 0.389$, $|\rho_{01}| = 0.358$, $F(|+\rangle) = 0.858$, $T_\varphi = 96\,\mu$s. Coherence-limited depth: $T_2 = 100\,\mu$s, 50 ns gates, coherence ≥ 0.9: 210 gates.



Repeated amplitude + phase damping steps reproduce e^{-t/T_1} and e^{-t/T_2} (CUDA-Q density matrix, illustrative).

Common trap

Scaling coherence by $1-\gamma$ instead of $\sqrt{1-\gamma}$; adding T's instead of rates; multiplying a T_2 decay by an extra T_1 factor.

7 Measurement errors

$\mathbf{q} = A\mathbf{p}$ with $A_{ij} = P(\text{read } i \,|\, \text{state } j)$. $P(1|0) = 0.02$, $P(0|1) = 0.05$: $|+\rangle$ is read as 1 with probability 0.485; a perfect 3-qubit GHZ shows population $\frac{1}{2}(0.98^3 + 0.95^3) = 0.899$. Mitigation: solve $A\mathbf{p} = \mathbf{q}$; e.g. $q_0 = 0.6$ gives $p_0 = 0.591$. Readout noise is *not* part of the notebook's noise model.

8 Coherent and stochastic errors

Key result: accumulation

Over-rotation ε per gate: $P(\text{error}) = \sin^2 \frac{n\varepsilon}{2}$ after n gates (quadratic). Stochastic flip with $p = \sin^2 \frac{\varepsilon}{2}$: $\frac{1}{2}[1-(1-2p)^n] \approx np$ (linear). $\varepsilon = 0.05$, $n = 60$: 0.995 vs. 0.036. Pauli twirling turns coherent errors into Pauli channels.

9 Simulating noise with CUDA-Q

method	target	properties
density matrix	`density-matrix-cpu`	exact ρ, any channel, $16 \cdot 4^N$ bytes ($N = 13$: 1.07 GB)
trajectories	`nvidia` + noise model	statistical, $16 \cdot 2^N$ per trajectory, GPU (not run here)
stabilizer	`stim`	Clifford + Pauli noise only, hundreds of qubits

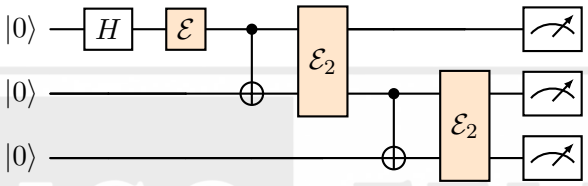
Common trap

(1) A constant `rx(0.0, q)` is removed by the compiler, so the noise attached to it never acts — pass the angle as an argument. (2) Keep a reference to the `NoiseModel` while CUDA-Q uses it (a temporary passed to `set_noise` crashed the process). (3) Noiseless targets ignore noise models. (4) `get_state` has no `noise_model` argument in CUDA-Q 0.16: use `set_noise`.

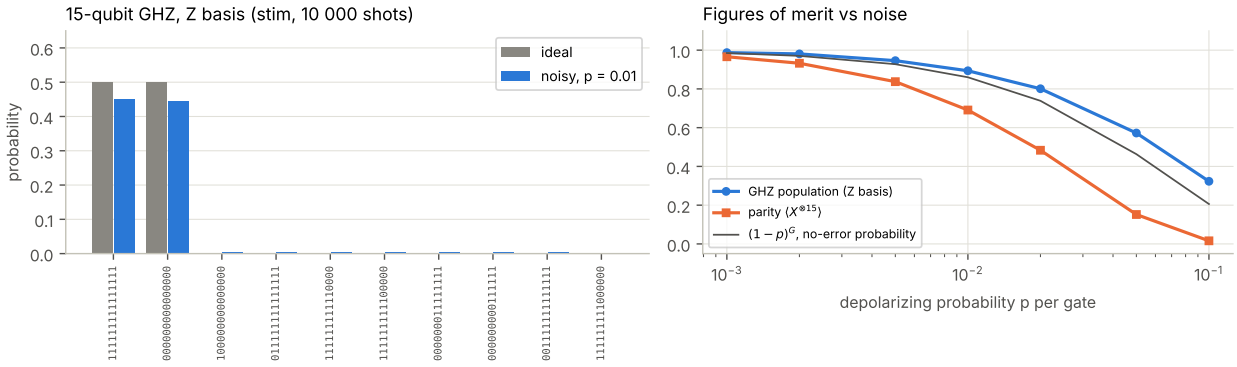
10 The notebook and its results

Definition: `build_noise_model(kind, p)`

Single-qubit channel after `h`, `x`, `ry`, `rz`; two-qubit channel after every CNOT (`num_controls=1`): `Depolarization2`, or `Pauli2` with $p(1-p), p(1-p), p^2$ (independent flips), or the product Kraus set for amplitude damping. `auto`: GPU → trajectories; CPU + GHZ + Pauli → `stim`; otherwise density matrix. Density-matrix counts are drawn from diag ρ with a seeded multinomial.



The notebook's GHZ circuit with its noise locations (3 qubits shown; the default has 15).



Re-executed default run: ideal vs. noisy distribution and figures of merit vs. p (identical to the recorded outputs).

Worked example 10.6: the recorded run (re-executed identically)

Depolarizing, `stim`: $p = 0.01$: population 0.8936, parity 0.6910, $(1-p)^{15} = 0.8601$, 96 outcomes; $p = 0.1$: 0.3232, 0.0160, 1165 outcomes. Leaked probability $1 - 0.8936 = 0.1064$. Bit-flip no-error probability $(1-p)^{29} = 0.7472$ at $p = 0.01$.

Key result: figures of merit

Hellinger $F_H = (\sum \sqrt{p_x q_x})^2$ (equal to the population for a balanced two-peak ideal), TVD $\frac{1}{2}\sum |p_x - q_x|$, parity $P(\text{even}) - P(\text{odd})$, state fidelity and purity (density matrix only). $(1-p)^G$ is a lower bound for Pauli noise, not the exact fidelity: 6-qubit GHZ at $p = 0.01$ has $F =$

$0.9492 > 0.99^6 = 0.9415$.

11 Fidelity measures and benchmarking

$F_{\text{avg}} = \frac{dF_{\text{pro}}+1}{d+1}$; depolarizing qubit: $F_{\text{pro}} = 1 - p$, $F_{\text{avg}} = 1 - \frac{2p}{3}$. Randomized benchmarking: $A\alpha^m + B$, $r = \frac{(d-1)(1-\alpha)}{d}$ ($\alpha = 0.99 \Rightarrow r = 0.005$). Both are theory only in this repository.

12 Error mitigation

Key result: zero-noise extrapolation

Linear $2E(1) - E(2)$; Richardson $3E(1) - 3E(2) + E(3)$; exponential $E(1)^2/E(2)$. Folding $G \to GG^\dagger G$: fold 4 of 10 gates $\Rightarrow \lambda = 1.8$. Lecture example (4-qubit GHZ parity, $p = 0.01\lambda$): $0.9055, 0.8189, 0.7396 \Rightarrow$ linear 0.9921, Richardson 0.9994, exponential 1.0021 (overshoot).

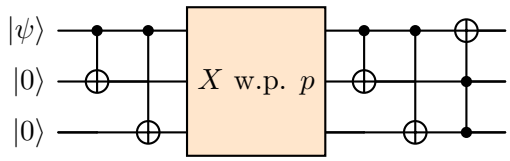
Common trap

ZNE and readout mitigation estimate expectation values — they do not repair the state, they cost extra shots, and they are model dependent.

13 Dynamical decoupling and error correction

Key result: echo

Static detuning: free decay $e^{-\sigma^2 t^2/2}$ ($\sigma = 1$, $t = 2$: 0.135); Hahn echo restores 1 because $XR_Z(a)X = R_Z(-a)$. Independent phase flips per step ($p = 0.02$, 20 steps): $0.96^{20} = 0.442$ with and without echo.



Encode, bit flips, decode and majority vote (Toffoli).

Key result: bit-flip code

$P_L = 3p^2 - 2p^3$ (0.028 at $p = 0.1$); useful for $p < \frac{1}{2}$. Surface code $p_L \propto (p/p_{\text{th}})^{(d+1)/2}$: $d = 3 \to 5$ at $p/p_{\text{th}} = 0.1$ gains $\times 10$.

14 Problem-solving strategies and question map

Method: five habits

1. **Convention first**: which depolarizing p? Which qubit first? Probability or amplitude?
2. **Go to the Bloch vector**: every single-qubit channel is an affine map of $\vec{r}$.
3. **Populations vs. coherences**: T_1/amplitude damping move populations; phase noise only coherences.
4. **Count locations**: one per gate for joint channels, two per CNOT for independent flips.
5. **Simulation $\neq$ mitigation $\neq$ correction $\neq$ hardware**.

Section	Assessment questions
1 Error statistics	7, 8, 13, 30, 31
2 Density matrices	1, 2, 3, 4, 12, 14, 15, 17, 18, 19, 20
3 Decoherence, fidelity	6, 10, 11, 32
4 Channels, CPTP	5, 16, 35, 43, 44, 45
5 Pauli channels	21, 22, 23, 33, 34, 36, 38, 41
6 Damping, T_1, T_2	24, 25, 26, 27, 28, 29, 37, 39, 40
7 Measurement errors	30, 31, 66, 67
8 Coherent vs. stochastic	9, 32, 72, 73
9 CUDA-Q simulation	41, 42, 48, 49, 53, 54, 55, 58, 74
10 Notebook and results	46, 47, 50, 51, 52, 56, 57, 59, 60, 78
11 Fidelity, benchmarking	61, 62
12 Error mitigation	63, 64, 65, 66, 67, 77, 79, 80
13 DD and QEC	68, 69, 70, 71, 75, 76

BRUCE FELLER

15 Practice problems

Try each before looking at the answers on the next page.

P1. Bloch vector and purity of $\left(\begin{smallmatrix} 0.75 & 0.25 \\ 0.25 & 0.25 \end{smallmatrix}\right)$.

P2. Depolarizing (CUDA-Q) $p = 0.15$ on $|+\rangle$: $\langle X \rangle$ and $P(-)$.

P3. Amplitude damping $\gamma = 0.19$ on $|+\rangle$: the output density matrix.

P4. The phase-flip probability equivalent to phase damping with $\lambda = 0.19$.

P5. $T_1 = 40\,\mu$s; a qubit starts in $|1\rangle$. $P(1)$ after $20\,\mu$s?

P6. $T_1 = 60\,\mu$s, $T_2 = 40\,\mu$s: T_φ?

P7. $P(1|0) = 0.03$, $P(0|1) = 0.08$: probability that $|11\rangle$ is read correctly.

P8. Bit flip $p = 0.25$ on $|0\rangle$: output and purity.

P9. Notebook GHZ with 10 qubits at $p = 0.02$: no-error probability for depolarizing and for bit-flip noise.

P10. Density-matrix memory for 11 qubits.

P11. ZNE: $E(1) = 0.90$, $E(2) = 0.82$. Linear estimate?

P12. Bit-flip code at $p = 0.2$: logical error.

P13. Readout calibration $P(0|0) = 0.97$, $P(0|1) = 0.06$; measured $q_0 = 0.5$. Mitigated p_0?

P14. Average gate fidelity of a depolarizing qubit channel with $p = 0.006$.

SILICO FELLER

Answers to the practice problems

P1. $\vec{r} = (0.5, 0, 0.5)$; purity $\frac{1+0.5}{2} = 0.75$.

P2. $\langle X \rangle = 1 - \frac{4(0.15)}{3} = 0.8$; $P(-) = 0.1$.

P3. $\left(\begin{smallmatrix} 0.595 & 0.45 \\ 0.45 & 0.405 \end{smallmatrix}\right)$ ($\rho_{00} = 0.5 + 0.19 \cdot 0.5$, $\rho_{01} = 0.9 \cdot 0.5$).

P4. $1 - 2p = \sqrt{0.81} = 0.9 \Rightarrow p = 0.05$.

P5. $e^{-0.5} = 0.607$.

P6. $\frac{1}{T_\varphi} = \frac{1}{40} - \frac{1}{120} = \frac{1}{60}$: $T_\varphi = 60\,\mu$s.

P7. $0.92^2 = 0.8464$.

P8. diag$(0.75, 0.25)$; purity $0.5625 + 0.0625 = 0.625$.

P9. Depolarizing $0.98^{10} = 0.817$; bit flip $0.98^{19} = 0.681$.

P10. $16 \cdot 4^{11} = 67.1$ MB.

P11. $2(0.90) - 0.82 = 0.98$.

P12. $3(0.04) - 2(0.008) = 0.104$.

P13. $p_0 = \frac{0.5-0.06}{0.97-0.06} = 0.484$.

P14. $1 - \frac{2(0.006)}{3} = 0.996$.

Summary of the chapter in ten lines

1. Errors multiply: $(1-p)^G$; shot noise $\sqrt{P(1-P)/S}$.
2. $\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$; purity $\frac{1+|\vec{r}|^2}{2}$; mixed $\Leftrightarrow$ inside the ball.
3. Decoherence = entanglement with the environment = random phases on average.
4. Channels: $\sum K \rho K^\dagger$, $\sum K^\dagger K = I$; Kraus sets not unique.
5. Bit/phase/bit-phase flip keep one axis; depolarizing shrinks by $1 - \frac{4p}{3}$.
6. Amplitude damping shifts towards $|0\rangle$ (T_1); phase damping $\leftrightarrow T_\varphi$; $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\varphi}$.
7. Readout errors: $\mathbf{q} = A\mathbf{p}$; coherent errors grow quadratically, stochastic linearly.
8. CUDA-Q: `stim` (Clifford + Pauli), density matrix (4^N), trajectories (2^N, GPU); keep the model alive; avoid constant identities.
9. The notebook's sweep is reproduced exactly; $(1-p)^G$ is a bound, parity reveals phase errors.
10. Mitigation estimates, DD refocuses slow noise, QEC protects — and none of this here ran on hardware.

SERGIO FELLER